

## Q.No 7

| DS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Session | Arrays | Question Information | Level 1 | Challenge 27 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|----------------------|---------|--------------|
| <p><b>Problem Description:</b></p> <p>Nairobi have given a matrix C of size N x M to Rio.</p> <p>Also Rio are given position of submatrix as X1, Y1 and X2, Y2 inside the matrix.</p> <p>Now Rio needs to find the sum of all elements inside that submatrix.</p> <p>Can you help Rio in completing the task assigned by Nairobi.</p> <p><b>Constraints:</b></p> <p><math>1 \leq T \leq 15</math></p> <p><math>1 \leq N, M \leq 103</math></p> <p><math>1 \leq C[N][M] \leq 106</math></p> <p><math>1 \leq X1, Y1, X2, Y2 \leq M</math></p> <p><b>Input Format:</b></p> <p>The first line of input contains an integer T denoting the number of test cases.</p> <p>The first line of each test case is n and m, n is the number of rows and m is the number of columns.</p> <p>The second line of each test case contains C[N][M].</p> <p>The third line contains four value of X1, Y1, X2, Y2. X1, Y1 is the top left cell and X2, Y2 is the bottom right cell.</p> |         |        |                      |         |              |
| <p><b>Output Format:</b></p> <p>Print the sum of all elements inside that submatrix.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |        |                      |         |              |
| <p>Logical Test Cases</p> <div> <div> <p><b>Test Case 1</b></p> <p>INPUT (STDIN)</p> <pre>1 3 3 9 18 7 14 2 11 6 51 3 1 2 1 3</pre> <p>EXPECTED OUTPUT</p> <p>79</p> </div> <div> <p><b>Test Case 2</b></p> <p>INPUT (STDIN)</p> <pre>1 3 3 9 18 7 14 2 11 6 51 3 1 2 1 3</pre> <p>EXPECTED OUTPUT</p> <p>25</p> </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |        |                      |         |              |

Code:

```
#include <stdio.h>
int main()
{
    int T;
    scanf("%d", &T);
    while(T--)
    {
        int n, m;
        scanf("%d %d", &n, &m);

        int a[n][m];

        int i, j;
```

```

for(i = 0; i < n; i++)
{
for(j = 0; j < m; j++)
{
scanf("%d", &a[i][j]);
}
}
int x1, y1, x2, y2;
scanf("%d %d %d %d", &x1, &y1, &x2, &y2);

int sum = 0;

for(i = x1 - 1; i <= x2 - 1; i++)
{
for(j = y1 - 1; j <= y2 - 1; j++)
{
sum = sum + a[i][j];
}
}

printf("output: %d\n", sum);
}

return 0;
}

```

Output:

Test\_case 1

```

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1
3 3
91 8 71 14 2 11 6 51 3
1 2 1 3
output: 79

```

Test\_case 2:

```

1
3 3
9 18 7 14 2 11 6 51 3
1 2 1 3
output: 25

```